Lead 5: Semaev S_3 in Ed25519's Edwards coordinate, F4 cost at m = 2

Date 2026-11-02. Branch `lead5-edwards-semaev`. Protocol: `PROTOCOL.md` (written before data) plus its Amendment 1 (pre-data: new CLI argument positions only).

Scope of every statement here: k = 20 miniature, one curve (no spread across curves), m = 2, t in {4,5,6,7}, hybrid guess G in {0,3}, intervals starting at 0, 100 targets per cell (two cells were cut short, see below), one seed per t. Nothing is said about m = 3, other offsets or sets, other k or curves, or the 255-bit setting. There is no complete DLP in this run, so no rho comparison is made.

What was built

- `PkmSemaevOracle::new_edwards_bits(m, t)` with `Membership::EdwardsBits`: the summands are Edwards y in [0, 2^t) written in bits, x_W = (1+y)/(1-y) + A/3, with S_3 cleared of denominators (`s3_in_u_frac`). The shift A/3 is derived in `prepare`. `from_spec` accepts `edwards_bits=T`. `edwards_factor_base` builds the matching base in the harness.
- `curve25519_family f4cost` form `edwards-bits`, with `max_nnz` as the last argument (position 12 after `[m] [mode] [lifts]`).
- Rebased onto the main line, keeping lead 2 (lifts), lead 3 (m = 3) and lead 4 (isogeny guessing).

Tests: `cargo test --release --lib pkm_semaev` gives `test result: ok. 14 passed; 0 failed; 0 ignored; 0 measured; 6075 filtered out`. The Edwards tests are `edwards_bits_form_agrees_with_grid` (F4 verdict equals grid verdict, with and without guessing, plus an explicit shift on a non-Montgomery curve), `edwards_bits_m3_planted_decompositions`, and `edwards_shift_is_the_two_torsion_abscissa`. `cargo clippy --release --lib --bin curve25519_family -- -D warnings` is clean (the only output is an unknown-lint warning for `clippy::chunks_exact_to_as_chunks`, which comes from the build flags).

Run

Frozen binary `/home/user/frozen-bin/curve25519_family_lead5`, sha256 in `analysis/binary.sha256` (`ec9cb40d...5f`). Command per cell (`run_cells.sh`): `curve25519_family f4cost 20 <form> <t> <G> 100 <2026111500+t> 600 2 full 0 30000000`, sequential, `ulimit -v 3500000`, `timeout 900`. Raw output: `cells/*.json`, `*.status` (with `uptime` at start and end), `*.stderr` (all empty). All 16 cells exited 0. The host was shared with other jobs; load average read 6.7 to 12.9 across the cell status files.

Per-cell table

`mul-adds/target` = F4 multiply-adds per target; `calls/target` = F4 calls per target (G = 3 makes 8 guess specialisations per target); `deg` = max solving degree; `hits/ver` = relations returned / re-summed to the target; `cov_exact` = the cell's `coverage_exact`; `per-rel` = mul-adds/target divided by `cov_exact` (`muladds_per_relation_at_exact_coverage`); `terms` = polynomial terms per specialisation; `wall` in seconds.

form	t	G	cols	targets	mul-adds/target	calls/target	deg	hits/ver	cov_exact	per-rel	capped	undecided	terms	wall
bits	4	0	7	100	4046085	1.0	6	0/0	9.1e-05	44238212441	no	0	121.0	1
edwards-bits	4	0	7	100	4033474	1.0	6	0/0	7.6e-05	52920398631	no	0	121.0	1
bits	4	3	7	100	88840	8.0	4	0/0	9.1e-05	971339543	no	0	22.0	0
edwards-bits	4	3	7	100	88839	8.0	4	0/0	7.6e-05	1165603122	no	0	22.0	0
bits	5	0	11	100	127284344	1.0	6	0/0	0.00032	397621138726	no	0	256.0	9

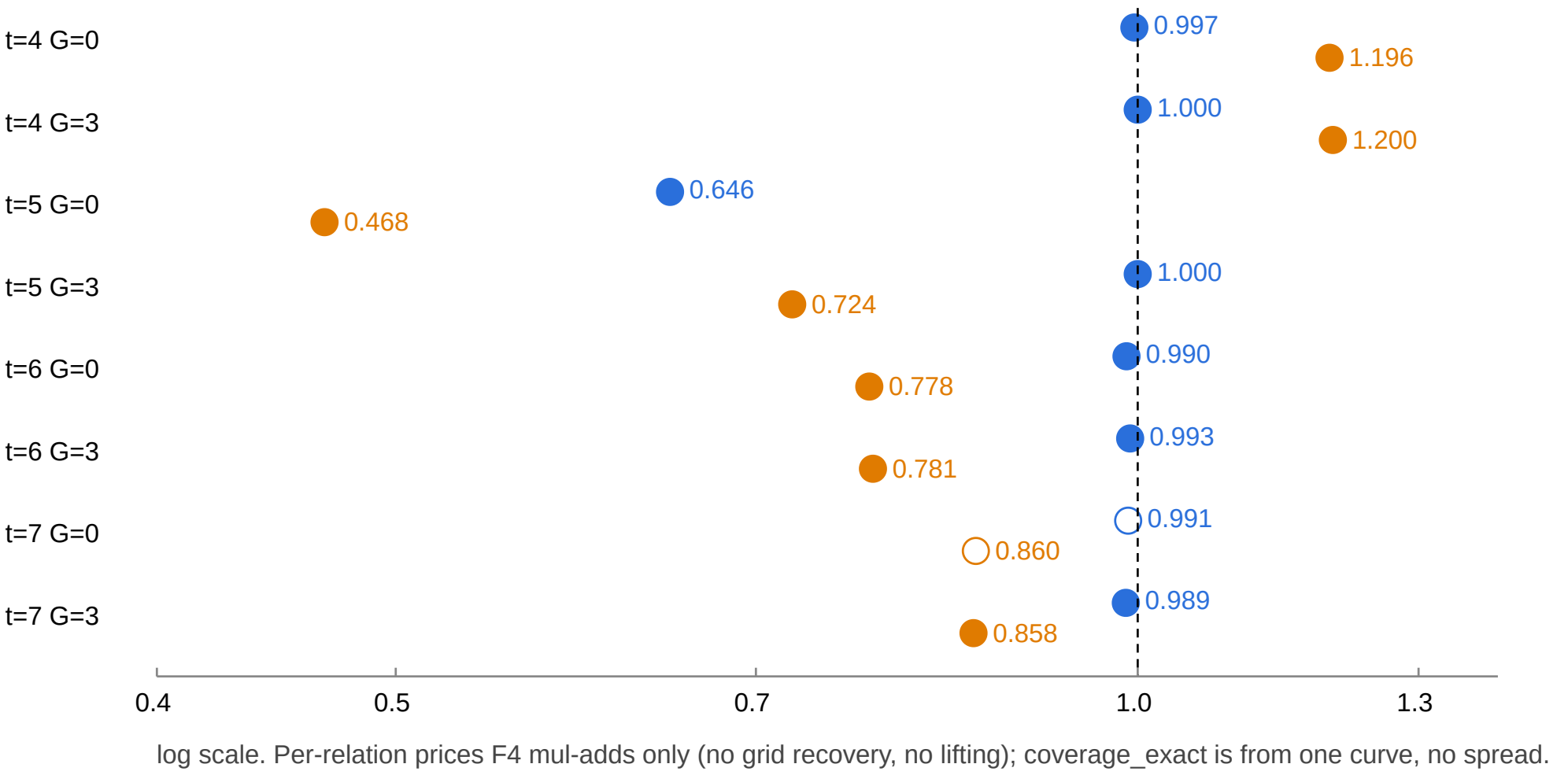
form	t	G	cols	targets	mul-adds/target	calls/target	deg	hits/ver	cov_exact	per-rel	capped	undecided	terms	wall
edwards-bits	5	0	14	100	82230218	1.0	6	0/0	0.000442	186014679579	no	0	256.0	7
bits	5	3	11	100	3036911	8.0	5	0/0	0.00032	9486950801	no	0	64.0	2
edwards-bits	5	3	14	100	3036922	8.0	5	0/0	0.000442	6869884812	no	0	64.0	1
bits	6	0	25	100	1753618663	1.0	7	0/0	0.001235	1420247095569	no	0	484.0	90
edwards-bits	6	0	29	100	1735358140	1.0	7	1/1	0.00157	1105263078493	no	0	484.0	84
bits	6	3	25	100	172425384	8.0	6	0/0	0.001235	139646467528	no	0	154.0	23
edwards-bits	6	3	29	100	171213569	7.99	6	1/1	0.00157	109047252082	no	0	154.0	26
bits	7	0	58	22	79397189460	1.0	7	0/0	0.00628	12642171661090	CAPPED (truncated)	0	841.0	601
edwards-bits	7	0	63	31	78692183259	1.0	7	0/0	0.007241	10868053179133	CAPPED (truncated)	0	841.0	616
bits	7	3	58	100	3484169705	8.0	6	0/0	0.00628	554773686790	no	0	319.0	204
edwards-bits	7	3	63	100	3445674737	7.93	6	0/0	0.007241	475876697388	no	0	319.0	201

The two t = 7, G = 0 cells hit the 600 s wall cap (22 targets for Weierstrass, 31 for Edwards, 601 s and 616 s) on a loaded host. **They are truncated samples**: their per-target means are over different, smaller target sets and are not comparable with the 100-target cells or with each other at full strength. No other cell was capped and no system was undecided.

Edwards vs Weierstrass, paired by (t, G)

Ratio = Edwards / Weierstrass. Per-relation uses each form's own `coverage_exact`.

t	G	mul-adds/target	per-relation (F4 mul-adds only)	note
4	0	0.997	1.196	
4	3	1.000	1.200	
5	0	0.646	0.468	Edwards base has 14 columns, Weierstrass 11
5	3	1.000	0.724	
6	0	0.990	0.778	
6	3	0.993	0.781	
7	0	0.991	0.860	both cells truncated (capped)
7	3	0.989	0.858	



ratios

What the measurements show, within the scope above:

- Solving degree is identical in every pair (6/6, 4/4, 6/6, 5/5, 7/7, 6/6, 7/7, 6/6), and so is the polynomial size per specialisation (121, 22, 256, 64, 484, 154, 841, 319).
- Mul-adds per target are within about 1.1 percent of each other in 7 of 8 pairs. The exception is t = 5, G = 0, where Edwards is 0.646 of Weierstrass; the same t with G = 3 shows a ratio of 1.000. The polynomial sizes are equal there, so this run does not say why G = 0 differs. It is a single cell with one seed.
- The per-relation ratio is mostly the ratio of `coverage_exact`: Edwards covers more points than Weierstrass at t >= 5 and fewer at t = 4 (cols 7 vs 7, cov_exact 7.6e-5 vs 9.1e-5). At t = 4 the per-relation figure is worse for Edwards, at t >= 5 it is better. The support figures behind this are single-curve counts (`analysis/support-cells.jsonl`, re-measured with the frozen binary), with no spread.

- Hits: 2 relations across the 16 cells, both from Edwards t = 6 cells (one at G = 0, one at G = 3, both re-summed to the target). Weierstrass returned none. With `coverage_exact` near 1e-3 and 100 targets, the expected number of hits per cell is about 0.1, so the hit counts do not discriminate between the forms.

Predictions (PROTOCOL.md, “Preregistered expectation”)

1. “Edwards S_3 has no lower solving degree at m = 2”: **held** in all 8 pairs (equal degree).
2. “Any F4 difference is expected to be a constant factor from sparsity and the number of points”: **inconclusive**. Seven pairs differ by at most about 1.1 percent in mul-adds per target, one pair (t = 5, G = 0) by a factor 0.646, with equal polynomial size, so sparsity of the cleared polynomial does not explain it here. The cause was not investigated. Different numbers of points show up in `coverage_exact`, which changes the per-relation figure.
3. The per-relation ordering (Edwards lower at t >= 5, higher at t = 4) was not predicted either way.

Deviations from the protocol

1. Amendment 1 (pre-data): CLI argument positions; cells run strictly sequentially instead of two at a time.
2. The test `edwards_bits_form_agrees_with_grid` was changed from t in {3,4,5} to {4,5}: on the Montgomery toy the t = 3 Edwards base has one column, which tripped the test’s own `columns >= 3` guard. The F4-versus-grid comparison is unchanged.
3. Per-relation cost uses each cell’s own `coverage_exact` field, not the `lambda` of the `support` subcommand. The two differ (for example t = 4: 9.1e-5 against 7.5e-4) and were not reconciled; `analysis/support-cells.jsonl` holds the support output as re-measured.
4. Two cells were truncated by the 600 s cap under load (above). Host load was not controlled; no repeat of any cell was run.
5. A first frozen-binary copy was another agent’s build from the shared target directory and was discarded before any cell ran; the binary used was rebuilt (forced recompile) and checked to contain this tree’s strings and to run `edwards-bits`.
6. A one-target smoke run of `edwards-bits` t = 4 was made to check the binary; it is not part of the data.

Not measured

Offsets or other windows, sets structured in y, m = 3 cost, other k, other curves, repeat seeds, spread between runs, the cost of grid recovery or lifting.